

Community Emergency Response App (CERA)

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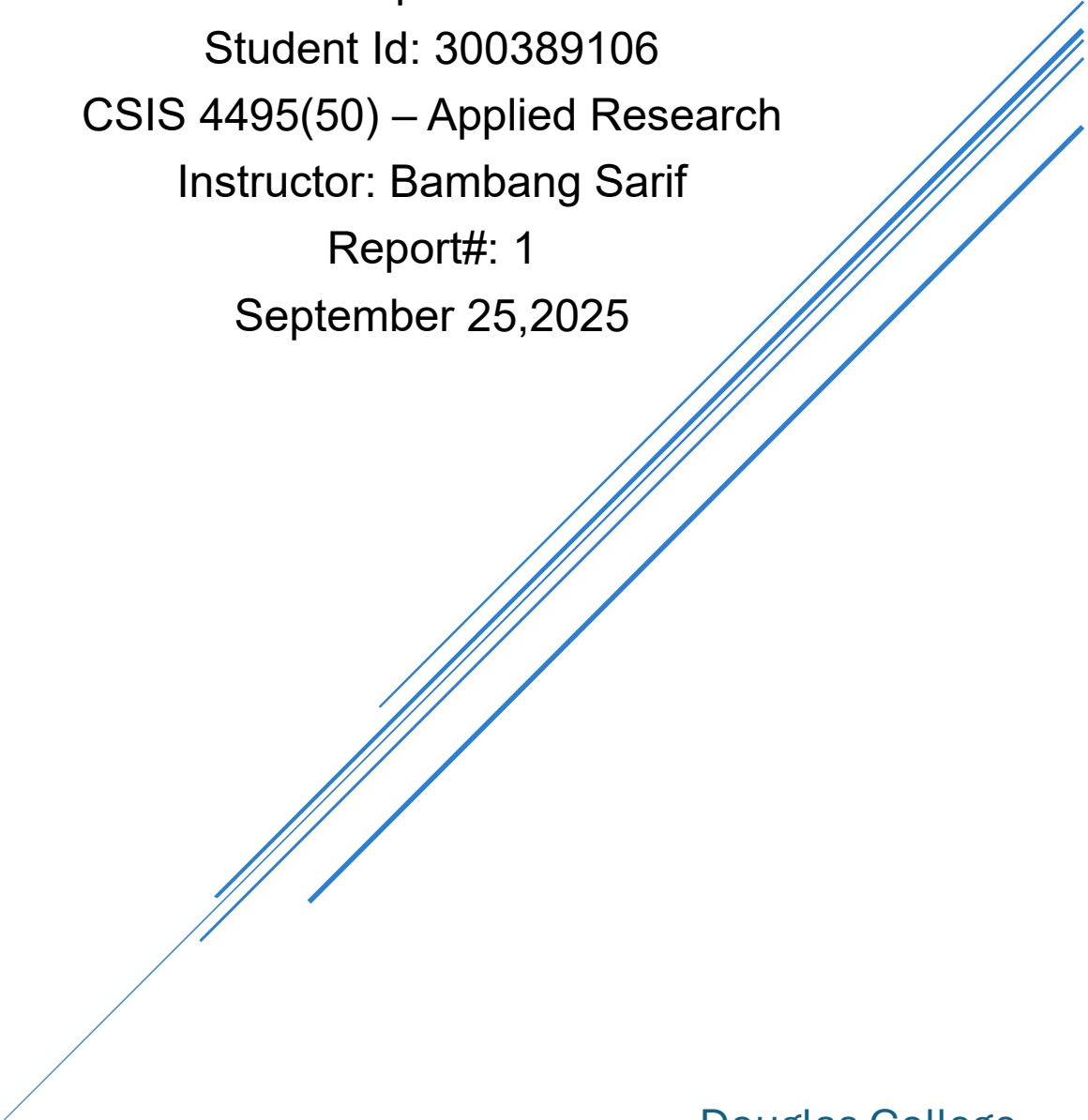
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CSIS 4495(50) – Applied Research

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Report#: 1

September 25, 2025

A series of five parallel blue lines of varying lengths, slanted diagonally from the bottom-left towards the top-right, positioned to the right of the student information.

Douglas College
CSIS 4495

Work Logs

Date	Number of Hours	Description of work done
Sep 13, 2025	2	Did a meeting on call with Harleen to get the update on the meeting with Riipen project owner. Moreover, decided to with our own project. Tried to finalise the topic for the app.
Sep 14, 2025	3	Initial research on emergency response apps. Added research notes for project proposal.
Sep 14, 2025	2.5	Finished meeting with Harleen and discussed on topics, but moved on with the emergency response app.
Sep 15, 25	2	Wrote implementation of Gantt Chart and User stories with Epics of them.
September 16, 25	3	Created draft for project set up on GitHub like folder, files, templates for issue, etc.
September 19, 25	5	Implemented the project draft by creating repo and adding project on GitHub, sent invite to collaborators, created Kanban board and some other modifications on GitHub.
September 21, 25	3	Created initial task for the project which are available on Kanban board, refer to attached screenshots at the bottom.
September 24, 25	5.5	Finished working on task of Epic 2, setting up FE, BE and database with some schema. However, some sub tasks are still in progress.
September 25, 25	4.5	Created several structure flows like ER diagrams, sequence diagrams, system architecture, API design, data relation diagram etc.

Description of Work done

Week 1 (Sept 4–Sept 11, 2025)

This week, I focused on forming the project team and exploring potential project ideas. On September 4, I created a project group with Harleen and reviewed several projects on Riipen, including “Glu” and “Weather Driver.” After evaluating the descriptions, we initially decided to go with Weather Driver. On September 11, I conducted further research on the project description and explored related applications to better understand the topic, functionality, and features of similar apps. This research helped in

gathering ideas for potential project implementation. The main challenge was choosing a project, which we solved by discussing and exploring different ideas.

Week 2 (Sept 13–Sept 15, 2025)

This week I worked on finalizing our project idea and starting the documentation. On September 13, I had a call with Harleen to understand the project requirements. After discussing the suggested projects, we decided to create our own app from scratch. We looked at several mobile and web applications for community services and volunteer coordination to get ideas and see how they work. On September 14, we settled on developing a Community Emergency Response App (CERA) and started writing the documentation, including the literature review, limitations, expected results, implications, and creating a Gantt chart. On September 15, I added detailed user stories to the Gantt chart, completing the planning and design phase.

Week 3 (Sept 18–Sept 25, 2025)

This week I worked on setting up the project foundation. The GitHub repo was structured with Implementation and Documents folders. I created the initial React Native app and backend skeleton with Express.js and verified MongoDB connectivity. Created tasks for Authentication flow and user schema definition for Users and Incidents. Moreover, I designed the app architecture, including sequence diagram, Work flow diagram, API design, data modeling etc.

Challenges included resolving the npm execution policies error in PowerShell(resolved by switching to CMD), git line ending issue(resolved by using global line-ending format for the project) and configuring MongoDB user roles.

Repo Check-In

The files/folders I have checked into the repo since the last report are as follows:

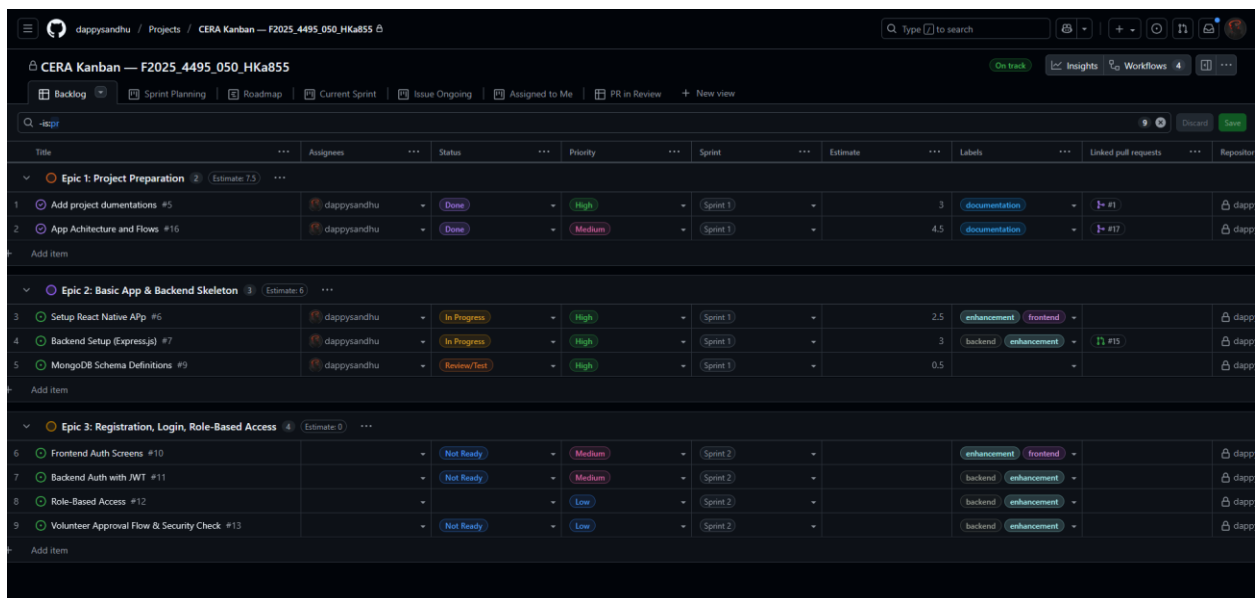
- Implementation/app/ → React Native base project
- Implementation/backend/ → Express.js server, routes, controllers
- Implementation/backend/models/ → User.js(sample)
- .github/workflows/ci.yml → GitHub Actions CI workflow
- Documents/ProjectProposal → initial project documentation

- Documents/CERA-architecture → project architectural and design documentation

Kanban Board

- **Epic 1:** Project Preparation → Task completed (*Add project documentation*).
- **Epic 2:** Basic App & Backend Skeleton → Tasks in progress (*React Native setup, Backend setup, MongoDB schema definitions*).
- **Epic 3:** Registration & Login → Tasks created but marked *Not Ready* for Sprint 2(2 weeks sprint).

Images from GitHub Project



Title	Assignees	Status	Priority	Sprint	Estimate	Labels	Linked pull requests	Repository
Epic 1: Project Preparation (Estimate: 7.5)								
1 Add project dumentations #5	dappysandhu	Done	High	Sprint 1	3	documentation	#1	dapp
2 App Architecture and Flows #16	dappysandhu	Done	Medium	Sprint 1	4.5	documentation	#17	dapp
Epic 2: Basic App & Backend Skeleton (Estimate: 6)								
3 Setup React Native App #6	dappysandhu	In Progress	High	Sprint 1	2.5	enhancement, frontend		dapp
4 Backend Setup (Express.js) #7	dappysandhu	In Progress	High	Sprint 1	3	backend, enhancement	#15	dapp
5 MongoDB Schema Definitions #9	dappysandhu	Review/Test	High	Sprint 1	0.5			dapp
Epic 3: Registration, Login, Role-Based Access (Estimate: 0)								
6 Frontend Auth Screens #10		Not Ready	Medium	Sprint 2		enhancement, frontend		dapp
7 Backend Auth with JWT #11		Not Ready	Medium	Sprint 2		backend, enhancement		dapp
8 Role-Based Access #12		Not Ready	Low	Sprint 2		backend, enhancement		dapp
9 Volunteer Approval Flow & Security Check #13		Not Ready	Low	Sprint 2		backend, enhancement		dapp

Figure 1: Kanban Board showing tasks in Backlog, In Progress, and Done.

Community Emergency Response App

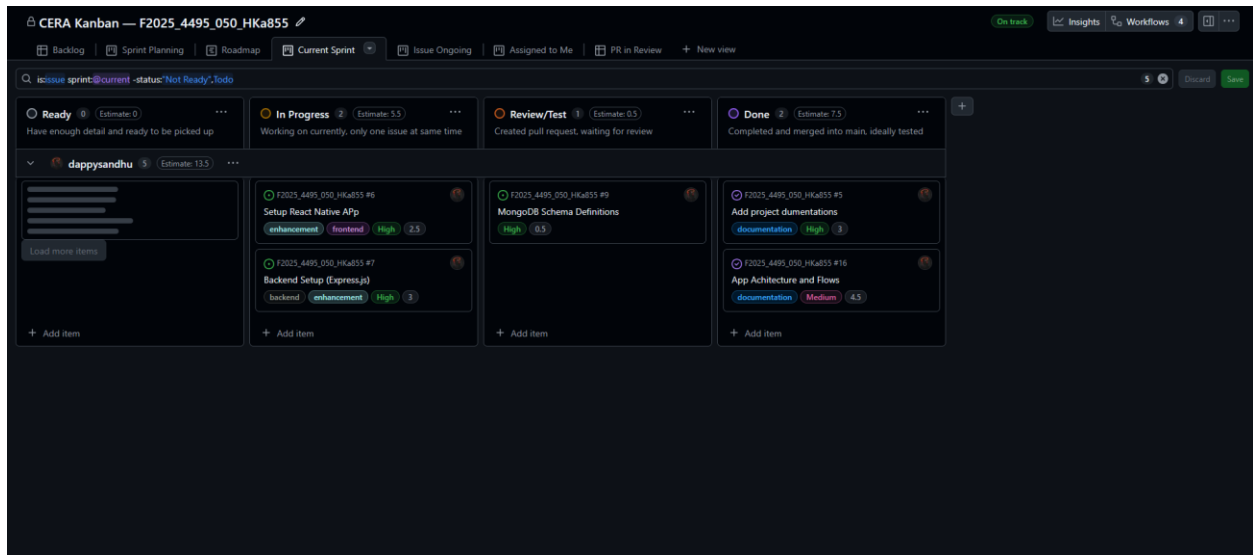


Figure 2: Current Ongoing Sprint.

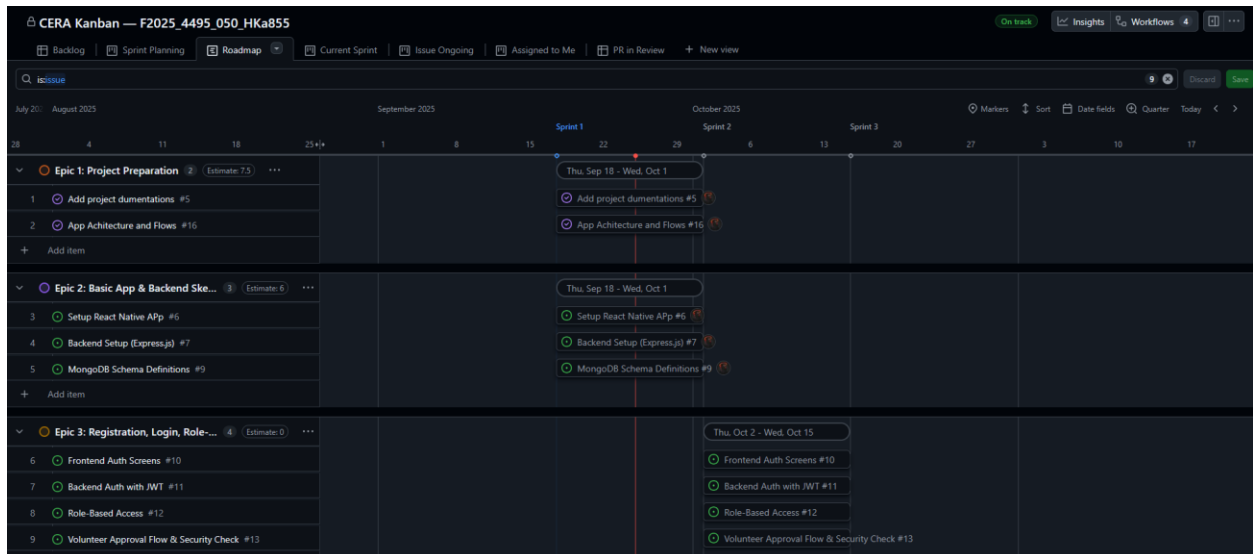


Figure 3: Project Roadmap.